

ACTEWAGL DISTRIBUTION 2016/17 STATEMENT OF TARIFF CLASSES AND TARIFFS

Effective date: June 2016

ActewAGL

for you

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Overview

- Electricity bills are made up of several components including the distribution costs (covering the poles and wires), transmission costs, ACT jurisdictional schemes,¹ metering and meter reading costs, and the retailer's costs and margins (covering wholesale energy costs) as well as GST. This report covers only the network components, which account for 20 to 30 per cent of an average electricity bill for residential and commercial consumers, respectively.
- ActewAGL Distribution offers customers a range of network tariff options across three tariff classes—residential, commercial low voltage and high voltage. Customers are able to choose the tariff that best suits their needs, subject to some eligibility requirements as set out in this document. No changes are proposed to the types of network tariffs offered in 2016/17.
- The 2016/17 tariffs and charges shown in this document have been set as per the Enforceable Undertaking (“Undertaking”) given by ActewAGL Distribution and accepted by the Australian Energy Regulator (AER) on 16 May 2016.²
- The proposed network use of system (NUOS) charges (comprising distribution, transmission, jurisdictional schemes and the capital component of metering) plus the non-capital component of metering for 2016/17 are, on average, 1.51 per cent higher in nominal terms than the average NUOS plus metering charges in 2015/16. The 1.51 per cent increase in prices in 2016/17 is based on CPI.
- ActewAGL Distribution estimates that the 2016/17 network and metering charges will increase the electricity network bill for a typical residential customer, consuming 7000 kWh on the Residential Basic Network tariff, by about \$0.19 a week (including GST).
- For a commercial customer consuming 30 MWh per annum on the General Network tariff, the network and metering charge would increase their network electricity bill by around \$1.09 per week (including GST).
- No changes are proposed to the structure of metering charges in 2016/17, following the significant changes introduced in 2015/16. From 1 July 2015, customers requesting new meters (for new connections or meter upgrades) have been required to pay the full AER approved cost for the meter at the time of installation. Customers who were connected before 1 July 2015 did not pay for their meter up-front. These customers will continue to pay an annual meter capital charge to contribute to the recovery of the cost of their meter.
- ActewAGL Distribution's miscellaneous charges (such as meter installations) will rise by between 2.2 per cent and 2.7 per cent in nominal terms in 2016/17 compared to 2015/16, as per the Enforceable Undertaking, and consistent with the AER's Final Decision.

¹ For example, the Energy Industry Levy, the Utilities Network Facilities Tax and the feed-in tariffs under the *Electricity Feed-in (Large-scale Renewable Energy Generation) Act 2011 (ACT)*.

² The Undertaking was supplied by ActewAGL Distribution because in February 2016 the Australian Competition Tribunal set aside the AER's 2015 distribution determination decision for ActewAGL Distribution. See <http://www.aer.gov.au/networks-pipelines/determinations-access-arrangements/pricing-proposals-tariffs/actewagl-annual-pricing-2016/17>

1. Introduction

1.1 Purpose and scope of the document

ActewAGL Distribution has prepared this statement of tariff classes and tariffs in accordance with the requirements in chapter 6 of the National Electricity Rules (the Rules).³ Clause 6.18.9 requires ActewAGL Distribution to maintain on its website:

- a statement of tariff classes⁴ and tariffs that are applicable to each class;
- for each tariff within a tariff class, the charging parameters⁵ and the elements of service to which they relate; and
- a statement of expected price trends, giving an indication of how ActewAGL Distribution expects prices to change over the regulatory period, and the reasons for those expected changes.

This document contains the required information. The statement covers the regulatory period (1 July 2015 to 30 June 2019). The prices for 1 July 2016 to 30 June 2017 are as published by the AER on 16 May 2016. The 2017/18 and 2018/19 prices (presented in Section 4) are indicative prices based on CPI assumptions approved by the AER in its Final Decision⁶.

1.2 Background

The AER is responsible for the economic regulation of distribution services provided by ActewAGL Distribution.

Following the release of the AER's Final Decision on 30 April 2015, ActewAGL Distribution applied to the Australian Competition Tribunal for merits review and the Federal Court for judicial review of the AER's final 2014–19 distribution determination. In February 2016, the Tribunal decided to set aside the AER's Final Decision. On 24 March 2016 the AER applied to the Federal Court for judicial review of the Australian Competition Tribunal decision to set aside the Final Decision⁷.

AAD acknowledged that having regard to the judicial review proceeding before the Federal Court and remittal process before the AER may cause significant delay in the AER remaking its decision with respect to AAD's distribution determination. The effect of this delay is likely to create uncertainty for users about applicable prices and the legal effect of the non-price provisions of the Final Determination. To ameliorate this uncertainty, AAD agreed to proffer an Undertaking to the AER⁸.

On 16 May 2016, the AER formally accepted ActewAGL Distribution's Undertaking that maintains network charges constant in real terms. The Undertaking will expire,

³ Under rule 11.73.1(b), the new chapter 6 pricing rules do not apply to ActewAGL Distribution until 1 July 2017. All references to Chapter 6 refer to the old Chapter 6.

⁴ A tariff class is defined in Chapter 10 of the *Network Electricity Rules* as "a class of for one or more direct control services who are subject to a particular tariff or particular tariffs."

⁵ Charging parameters are defined as "the constituent elements of a tariff" in chapter 10 of the *National Electricity Rules*.

⁶ AER 2015, *Final Decision, ActewAGL distribution determination 2015/16 to 2018/19*.

⁷ <http://www.aer.gov.au/networks-pipelines/determinations-access-arrangements/actewaggl-determination-2014-19>

⁸ <http://www.aer.gov.au/system/files/ActewAGL%20enforceable%20undertaking%20-%20May%202016.pdf>

at the latest, on 30 June 2017. The Undertaking obliges ActewAGL Distribution to continue to provide network services consistent with the non-price terms and conditions of its 2015–19 electricity distribution determination⁹.

This document should be read in conjunction with the following documents.

- the AER's Final Decision
- ActewAGL Distribution's Subsequent Regulatory Proposal
- ActewAGL Distribution's Revised Regulatory Proposal
- ActewAGL Distribution's Enforceable Undertaking – May 2016

These documents are published on the AER's website, and set out in detail the basis of the costs that are reflected in ActewAGL Distribution's prices.

1.3 Structure of the document

ActewAGL Distribution's tariff structure and 2016/17 charges for network services are set out in chapter 2. The chapter includes details of the components and rationale for each tariff, as well as a discussion of the changes relative to 2015/16.

The structure and basis of ActewAGL Distribution's charges for ancillary network services and metering services, the charges for 2016/17 and the changes relative to 2015/16 are presented and explained in chapter 3.

Expected price trends over the period 2017/18 to 2018/19 are provided in chapter 4.

2. Network tariffs

The Rules (clause 6.18.9) require a description of the *tariff classes*¹⁰ and the tariffs that apply to each class in 2016/17, to be published on ActewAGL Distribution's website.

2.1 Network tariff structure

ActewAGL Distribution offers network tariffs in three tariff classes:

- Residential;
- Commercial low voltage (LV); and
- High voltage (HV).

The Rules stipulate that tariff classes must be constituted with regard to the need to group customers together on an economically efficient basis and the need to avoid unnecessary transactions costs (clause 6.18.3(d)). ActewAGL Distribution meets this requirement by grouping customers according to type of connection (residential or commercial), and connection voltage (LV or HV). Customers within each class

⁹ <http://www.aer.gov.au/networks-pipelines/determinations-access-arrangements/pricing-proposals-tariffs/actewagl-annual-pricing-2016/17>

¹⁰ A tariff class is defined in chapter 10 of the *National Electricity Rules* as "a class of customers for one or more direct control services who are subject to a particular tariff or particular tariffs".

have similar load and connection characteristics. The relevant costs for each class can then be identified and reflected in the tariffs for each class.

Within each of these three tariff classes, ActewAGL Distribution has developed a suite of network tariffs that effectively meet the diverse needs of its customer base, encourage efficient use of the network and signal the costs of future network expansion. Residential customers are offered a choice of four network tariff options plus two controlled load off-peak options and an embedded renewable generation tariff. Commercial LV customers are offered four main tariff options. Commercial customers on the general network tariff also have access to the controlled load off-peak tariff options and the embedded renewable generation tariff on a similar basis to customers in the residential class. Commercial HV customers are offered four tariff options. Customers are able to choose the option which best suits their needs, subject to the eligibility criteria set out in Tables 2.1 to 2.3 below.

The network tariffs comprise different combinations of the following *charging parameters*:

- Network access charges—these apply per customer for residential consumers and per connection point for commercial customers. They involve a fixed daily charge and do not vary with electricity consumption or capacity;
- Energy charges—these apply to each unit of electricity consumed. The c/kWh rate may vary with the level of consumption (with higher rates applying above certain thresholds) or with the time-of-use (with lower rates applying at off-peak periods);
- Maximum demand charges—these apply per connection point for some commercial tariffs. They involve a charge per unit of maximum demand (in c/kVA/day). The maximum demand is the highest demand calculated over a 30-minute interval during the billing period;
- Capacity charges—these apply on the same basis as maximum demand charges, but are for the maximum demand calculated over a 30-minute interval during the previous 13 months.

Network access charges relate to the connection services provided to customers. They are associated with the costs of the provision and use of connection assets for each tariff class, as well as customer related costs such as network call centre costs.

Energy charges relate to the distribution services provided to customers. They are linked to the costs of the provision and use of distribution assets (other than connection assets), and also recover most of the common services costs. Higher energy rates at peak periods reflect higher costs. Higher energy rates beyond 330 kWh per day for the general network charge encourage larger consumers with a good load factor to move to demand or time-of-use network charges.

Maximum demand and capacity charges relate to the capacity provided to meet maximum demand and provide incentives for consumers to manage their load on the network.

The allocation of costs to charging parameters is discussed further in section 2.4 below.

The tariffs and charging parameters for each tariff class are shown in the following tables (2-1 to 2-3). The tables include an explanation of the purpose of each tariff and the customers to which each tariff may apply.

2.1.1 Network tariffs for residential customers

ActewAGL Distribution's residential network tariff structure is shown in Table 2-1. The *Residential time-of-use (TOU)*, *Residential 5000* and *Residential with heat pump* tariffs are refinements of the *Residential basic* tariff to reflect customer load profiles. The *Residential TOU* tariff provides an opportunity and an incentive for customers with the necessary metering capability to respond to price signals at different times of the day, where reflected in the final price of their retailer, and manage their electricity bill in line with the costs they impose on the network. The Residential TOU tariff is the default tariff for all new residential connections. The *Residential 5000* and *Residential with heat pump* tariffs involve a higher connection charge and an inclining block structure with a higher energy charge (cents per kWh) applying above certain thresholds. These tariffs more accurately tailor costs to the load profile of these customers. The off-peak tariff options can be used in conjunction with the *Residential basic* and the *Residential TOU* network tariffs.

Table 2-1 Network tariff structure – residential

<i>Tariff</i>	<i>Charging parameters</i>	<i>Explanation</i>
Residential basic network	Network access charge (c/day/customer) Energy charge (c/kWh)	The residential basic network tariff is available to installations at private dwellings, excluding serviced apartments, but including: • Living quarters for members and staff of religious orders; • Living quarters on farms; • Charitable homes; • Retirement villages; • Residential sections of nursing homes and hospitals; • Churches, buildings or premises which are primarily used for public worship; and • Approved caravan sites. The energy charge varies neither with the level of consumption nor the time of day. However, customers on this tariff are also eligible for the off-peak tariffs.
Residential time-of-use (TOU) network ¹¹	Network access charge (c/day/customer) Energy at max times, ie 7am to 9am and 5pm to 8pm every day (c/kWh) Energy at mid times, ie 9am to 5pm and 8pm to 10pm every day (c/kWh) Energy at economy times, ie all other times (c/kWh)	This tariff is available to residential customers (as defined above) with a meter able to be read as a time-of-use meter and to electric vehicle recharge facilities on residential premises. The energy charges relate to the supply of network services at various times. Higher rates apply at max or peak times to encourage users to shift their load to off-peak periods. Customers on this tariff are also eligible for the controlled load off-peak tariffs. Residential consumers with a meter with two registers capable of providing time-of-use consumption data from each register may have the time-of-use charges applied separately to each register.
Residential 5000 network	Network access charge (c/day/customer) Energy for the first 60 kWh/day (c/kWh) Energy above 60 kWh/day (c/kWh)	This tariff is designed for residential customers who have large continuous (rather than time controlled) loads, such as electric hot water systems, and consume over 5,000 kWh per annum. The energy charges relate to the supply of network services above and below certain volume thresholds. An inclining block structure applies (ie higher energy rates for the second block of energy). The lower energy rate is limited to consumption up to 60 kWh per day, reflecting a typical domestic usage profile. This is sufficient to cover the energy requirements of many residential consumers.
Residential with heat pump	Network access charge (c/day/customer) Energy for the first 165 kWh/day (c/kWh) Energy above 165 kWh (c/kWh)	This tariff is only available to residential customers with a reverse cycle air conditioner. An inclining block structure applies (ie higher energy rates for the second block of energy). The lower energy rate is set to recover the incremental cost of energy load on the network as a demand management tool to lower winter peak loads and improve utilisation of the network in summer and so improve overall network utilisation.
Off-peak (1) night network	Energy at controlled times, ie between 10 pm and 7 am (c/kWh)	The off-peak (1) night charge is available only to customers utilising a controlled load element, and taking all other energy at residential basic network, residential time-off-use or general network rates. The off-peak (1) night charge is applicable to permanent heat (or cold)

¹¹ All times for metering are Eastern Standard Time.

<i>Tariff</i>	<i>Charging parameters</i>	<i>Explanation</i>
		storage; electric vehicle recharge; and CNG vehicle gas compression installations. The design and rating must be acceptable to ActewAGL. The installation must use most energy during the controlled times but may be boosted at the principal charge, or charges, at other times. The off-peak (1) night network energy charge relates to supply of network services at controlled times, for 6 to 8 hours per day between the hours of 10 pm and 7 am.
Off-peak (3) day and night network	Energy at controlled times, ie between 10 pm and 7 am and 9 am and 5 pm (c/kWh)	Available only to customers utilising a controlled load element, and taking all other energy at residential basic network, residential time-off-use or general network rates. This charge is applicable to permanent heat (or cold) storage installations. The design and rating must be acceptable to ActewAGL. The off-peak (3) day and night network energy rate applies to power supplied for up to 13 hours per day between 10 pm and 7 am and again between 9 am and 5 pm.
Renewable generation	Energy charges (c/kWh)	This tariff applies to customers with grid connected solar or wind energy generation systems. Different arrangements apply to customers participating in the ACT feed-in tariff scheme, in accordance with the Electricity Feed-in (Renewable Energy Premium) Act 2008 (ACT). Net metering applies to new PV installations since July 2013.

For each of the tariffs shown in this table (other than off-peak and renewable energy) two separate codes will apply – one which includes a meter capital charge and one which excludes the meter capital charge (XMC). The basis for the separate meter capital charges is explained in section 3.2 below.

2.1.2 Network tariffs for low voltage commercial customers

ActewAGL Distribution sets different tariffs for commercial low voltage (LV) and commercial high voltage (HV) customers recognising the different costs associated with supplying each group. Within the commercial LV tariff class a range of tariff options has been developed to meet the diverse needs of commercial customers and to accommodate their differing load profiles and ability to respond to price signals. Of the four main options offered to commercial LV customers, all but the *General network* tariff involve time-of-use charges. The *General network* tariff does, however, involve an inclining block tariff structure with higher energy charge (c/kWh) applying above a threshold of 330 kWh per day (about 10 MWh per month). Also, the off-peak (controlled load) tariffs are available to customers on the *General network* tariff.

Two of the commercial LV options involve capacity and/or maximum demand charges, in conjunction with time-of-use energy charges. Customers able to improve their load factor¹² have an incentive to choose a tariff with a demand or capacity component, to reduce their electricity bills. Customers on the General network and General time-of-use network charges can move to the demand tariffs and lower

¹² The load factor is the ratio of average load to peak load (maximum demand).

their network costs if they have a sufficiently large load (for the cost savings to offset the higher cost of interval metering) and a reasonable load profile (consumption is manageable during peak times).

Table 2-2 Network tariff structure - commercial low voltage

<i>Tariff</i>	<i>Charging parameters</i>	<i>Explanation</i>
General network	Network access charge (c/day/customer) Energy for the first 330 kWh/day (c/kWh) Energy above 330 kWh/day (c/kWh)	This tariff is available to all customers. The tariff is most suitable for small commercial consumers operating in regular business hours or larger customers with poorer load factors (peaky loads). This tariff may be used in conjunction with the off-peak tariffs.
General TOU network	Network access charge (c/day/customer) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is particularly suitable for small commercial customers with discretionary or relatively large off-peak loads such as bakers, freezer installations and irrigators. The energy charges relate to supply of network services at different times.
LV TOU kVA demand network	Network access charge (c/day/connection point). Maximum demand (in billing period) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is appropriate for customers with an average or stable commercial load. The maximum demand charge is designed to encourage consumers to manage their demand upon the network. The energy charges relate to supply of energy at different times, with lower rates in off-peak times reflecting the availability of capacity and encouraging consumers to shift their load from peak to off-peak times to utilise the available capacity.
LV TOU capacity network	Network access charge (c/day/connection point) Maximum demand (in billing period) (c/kVA/day) Capacity (max demand in last year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is open to all low voltage consumers and intended to reward those customers with seasonally stable loads. It is prescribed for low voltage customers with embedded generation. The tariff provides an incentive for customers with embedded generation to manage their output and their down-times (eg for servicing) so as to minimise their demand on the network.
Streetlighting	Network access charge (c/day/customer) Energy at any time (c/kWh)	This tariff applies to the night-time lighting of streets and public ways and places.
Small unmetered loads	Network access charge (c/day/customer) Energy at any time (c/kWh)	This tariff applies to eligible installations as determined by ActewAGL, including: • telephone boxes • telecommunication devices • other, as determined by the National Metrology Coordinator. Energy charges are calculated based on the assessed rating of the load and the charge period.

* Business times are between 7 am and 5 pm Eastern Standard Time on weekdays. Evening times are between 5 pm and 10 pm Eastern Standard Time on weekdays. Off-peak times are all other times.

For each of the tariffs shown in this table (except small unmetered loads), two separate codes will apply – one which includes a meter capital charge and one which excludes the meter capital charge (XMC). The basis for the separate meter capital charges is explained in section 3.2 below.

2.2 Network tariffs for high voltage customers

To qualify for the high voltage demand network charges, consumers must take their energy at high voltage (nominal voltage not less than 11 kV) and make a capital contribution towards their connection assets and transformers. High voltage consumers have the option of owning and operating their own high voltage assets. Some customers have aggregated their load, incorporating part of ActewAGL Distribution's low voltage network to become a high voltage customer. A separate high voltage network charge is available for such customers.

Customers taking their energy at high voltage also have the option of selecting the network tariffs available to low voltage consumers. For example, a high voltage customer with a poor load factor could select the *General* or the *General time-of-use* network charge.

Table 2-3 Network tariff structure - high voltage

<i>Tariff (code)</i>	<i>Charging parameters</i>	<i>Explanation</i>
HV TOU Demand Network (111)	Network access charge (c/day/connection point) Max demand (in billing period) (c/kVA/day) Capacity (max demand in past year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is appropriate for large customers taking supply at high voltage with a low voltage network owned and maintained by ActewAGL. The energy charges relate to supply of network services at different times, with lower rates in off-peak times encouraging customers to increase their utilisation of the network in off-peak periods. The demand charge is applied to the maximum demand in the billing period while the capacity charge is applied to the maximum demand in the previous 12 months. The capacity charge encourages consumers to monitor and manage their peak demand over the year while the demand charge continues to encourage consumers to manage their demand requirements each month.
HV TOU Demand Network – Customer HV (112)	Network access charge (c/day/connection point) Max demand (in billing period) (c/kVA/day) Capacity (max demand in past year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is appropriate for large customers taking supply at high voltage with a low voltage network owned and maintained by ActewAGL, where the customer owns and is responsible for their high voltage assets (including transformers and switch gear).
HV TOU Demand Network – Customer LV (121)	Network access charge (c/day/connection point) Max demand (in billing period) (c/kVA/day) Capacity (max demand in past year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This network charge is appropriate for large customers taking supply at high voltage where the customer owns and is fully responsible for their own low voltage network. The network access charge relates to the connection services provided to the customer including provision of the current transformer necessary to meter these large loads. The capacity charge encourages the consumer to monitor and manage their peak demand over the year while the demand charge continues to encourage consumers to manage their capacity requirements each month. The energy charges relate to supply of network services at different times, with lower rates in off-peak times reflecting the relatively low costs of off-peak supply, and thereby providing incentives for customers to switch their utilisation of the network to off-peak periods.
HV TOU Demand Network – Customer HV and LV (122)	Network access charge (c/day/connection point) Max demand (in billing period) (c/kVA/day) Capacity (max demand in past year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This network charge is appropriate for large customers taking supply at high voltage where the customer owns and is fully responsible for their own low voltage network and where the customer owns and is responsible for their high voltage assets (including transformers and switch gear).

* Business times are between 7 am and 5 pm Eastern Standard Time on weekdays. Evening times are between 5 pm and 10 pm Eastern Standard Time on weekdays. Off-peak times are all other times.

2.2.1 Ancillary network charges

In addition to the network tariffs set out above, ActewAGL Distribution offers a range of ancillary network services. The structure of each ancillary service charge depends on the type of service. Some services are charged on a per visit basis, others per installation or per test. The charges for ancillary network services are set on a cost reflective basis, in accordance with the AER's Final Decision¹³. For example, separate rates apply for temporary connections depending on whether they relate to an overhead or underground connection, as these will involve different costs. Ancillary network services and metering services charges are discussed further in chapter 3.

2.3 Network charges for 2016/17

ActewAGL Distribution's network charges for 2016/17 are made up of distribution charges, transmission charges, jurisdictional scheme charges and metering capital charges. The distribution charges recover the cost of ActewAGL's electricity distribution service within the ACT. The transmission charges recover TransGrid's charges to ActewAGL for the delivery of energy to the ACT and most of ActewAGL Distribution's costs for sub-transmission services.

The AER regulates both ActewAGL's distribution charges and TransGrid's transmission charges. Also, the National Electricity Rules (NER) provide that ActewAGL Distribution is to recover the cost of jurisdictional schemes including the ACT feed-in tariff, the Utilities Network Facilities Tax and the Energy Industry Levy in its network charges.

Furthermore, the AER in its Final Decision required ActewAGL Distribution to charge new customers (after 1 July 2015) the full cost of their meter. To facilitate that change, the AER split metering charges into two components: one to recover the capital costs and the other to recover metering operating and maintenance costs. To apply these charges in the manner that the AER has determined, ActewAGL Distribution included the capital metering cost in the network charges for existing customers (that is, customers with regulated meters installed before 30 June 2015). New customers that have paid for their meter are placed on network charges that exclude metering capital charges (XMC tariffs) and pay only the non-capital component of the metering charge. Table 2-4 shows the AER approved distribution, transmission, jurisdictional scheme, metering capital and network charges for 2016/17, excluding GST.

¹³ AER 2015, *Final Decision, ActewAGL distribution determination 2015/16 to 2018/19*, Attachment 16

Table 2-4 Network use of system charges 2016/17 (excluding GST)

		Distributi on Charges	Transmissi on Charges	Jurisdiction al Charges	Metering Capital	Network Charges
Description	Unit	2016/17	2016/17	2016/17	2016/17	2016/17
RESIDENTIAL TARIFFS						
10 Residential Basic Network						
Network access charge	cents/day	25.641	0.000	0.000	7.644	33.285
Energy consumption	cents/kWh	3.485	2.385	1.032		6.902
11 Residential Basic Network XMC*						
Network access charge	cents/day	25.641	0.000	0.000		25.641
Energy consumption	cents/kWh	3.485	2.385	1.032		6.902
15 Residential TOU Network						
Network access charge	cents/day	25.641	0.000	0.000	7.644	33.285
Energy at max times	cents/kWh	9.163	1.484	1.290		11.937
Energy at mid times	cents/kWh	3.716	1.018	1.032		5.766
Energy at economy times	cents/kWh	1.126	0.715	0.767		2.608
16 Residential TOU Network XMC						
Network access charge	cents/day	25.641	0.000	0.000		25.641
Energy at max times	cents/kWh	9.163	1.484	1.290		11.937
Energy at mid times	cents/kWh	3.716	1.018	1.032		5.766
Energy at economy times	cents/kWh	1.126	0.715	0.767		2.608
20 Residential 5000 Network						
Network access charge	cents/day	47.161	0.000	0.000	7.644	54.805
Energy for the first 60 kWh per day	cents/kWh	2.181	2.147	1.032		5.360
Energy above 60 kWh per day	cents/kWh	3.485	2.385	1.032		6.902
21 Residential 5000 Network XMC						
Network access charge	cents/day	47.161	0.000	0.000		47.161
Energy for the first 60 kWh per day	cents/kWh	2.181	2.147	1.032		5.360
Energy above 60 kWh per day	cents/kWh	3.485	2.385	1.032		6.902
30 Residential with Heat Pump Network						
Network access charge	cents/day	90.505	0.000	0.000	7.644	98.149
Energy for the first 165 kWh per day	cents/kWh	0.724	2.132	1.032		3.888
Energy above 165 kWh per day	cents/kWh	3.485	2.385	1.032		6.902
31 Residential with Heat Pump Network XMC						
Network access charge	cents/day	90.505	0.000	0.000		90.505
Energy for the first 165 kWh per day	cents/kWh	0.724	2.132	1.032		3.888
Energy above 165 kWh per day	cents/kWh	3.485	2.385	1.032		6.902
60 Off-Peak (1) Night Network						
Energy consumption	cents/kWh	0.076	1.188	0.614		1.878
70 Off-Peak (3) Day & Night Network						
Energy consumption	cents/kWh	0.351	1.653	0.767		2.771

COMMERCIAL LOW VOLTAGE TARIFFS

40 General Network

Network access charge	cents/day	47.039	0.000	0.000	13.369	60.408
Energy for the first 330 kWh per day	cents/kWh	6.353	3.344	1.032		10.729
Energy above 330 kWh per day	cents/kWh	9.546	3.379	1.032		13.957

41 General Network XMC

Network access charge	cents/day	47.039	0.000	0.000		47.039
Energy for the first 330 kWh per day	cents/kWh	6.353	3.344	1.032		10.729
Energy above 330 kWh per day	cents/kWh	9.546	3.379	1.032		13.957

135 Small Unmetered Loads Network

Network access charge	cents/day	38.269	0.000	0.000		38.269
Energy consumption	cents/kWh	9.276	1.211	0.720		11.207

80 Streetlighting Network

Network access charge	cents/day	47.536	0.000	0.000	13.369	60.905
Energy consumption	cents/kWh	5.305	0.971	1.043		7.319

81 Streetlighting Network XMC

Network access charge	cents/day	47.536	0.000	0.000		47.536
Energy consumption	cents/kWh	5.305	0.971	1.043		7.319

90 General TOU Network

Network access charge	cents/day	47.039	0.000	0.000	13.369	60.408
Energy at business times	cents/kWh	13.399	2.199	1.364		16.962
Energy at evening times	cents/kWh	6.149	1.142	1.032		8.323
Energy at off-peak times	cents/kWh	2.483	0.219	0.739		3.441

91 General TOU Network XMC

Network access charge	cents/day	47.039	0.000	0.000		47.039
Energy at business times	cents/kWh	13.399	2.199	1.364		16.962
Energy at evening times	cents/kWh	6.149	1.142	1.032		8.323
Energy at off-peak times	cents/kWh	2.483	0.219	0.739		3.441

Low voltage time of use demand network

101 LV TOU kVA Demand Network

Network access per connection point	cents/day	51.059	0.000	0.000	107.903	158.962
Maximum demand charge	c/KVA/day	29.263	13.066	0.000		42.329
Energy at business times	cents/kWh	2.581	2.582	1.364		6.527
Energy at evening times	cents/kWh	1.791	0.659	1.032		3.482
Energy at off-peak times	cents/kWh	0.786	0.038	0.739		1.563

103 LV TOU Capacity Network

Network access per connection point	cents/day	51.059	0.000	0.000	107.903	158.962
Maximum demand charge	c/KVA/day	16.729	3.065	0.000		19.794
Capacity charge	c/KVA/day	16.729	3.065	0.000		19.794
Energy at business times	cents/kWh	3.885	1.278	1.364		6.527
Energy at evening times	cents/kWh	2.129	0.320	1.032		3.481
Energy at off-peak times	cents/kWh	0.809	0.015	0.739		1.563

104 LV TOU kVA Demand Network XMC

Network access per connection point	cents/day	51.059	0.000	0.000	51.059
Maximum demand charge	c/KVA/day	29.263	13.066	0.000	42.329
Energy at business times	cents/kWh	2.581	2.582	1.364	6.527
Energy at evening times	cents/kWh	1.791	0.659	1.032	3.482
Energy at off-peak times	cents/kWh	0.786	0.038	0.739	1.563

105 LV TOU Capacity Network XMC

Network access per connection point	cents/day	51.059	0.000	0.000	51.059
Maximum demand charge	c/KVA/day	16.729	3.065	0.000	19.794
Capacity charge	c/KVA/day	16.729	3.065	0.000	19.794
Energy at business times	cents/kWh	3.885	1.278	1.364	6.527
Energy at evening times	cents/kWh	2.129	0.320	1.032	3.481
Energy at off-peak times	cents/kWh	0.809	0.015	0.739	1.563

HIGH VOLTAGE TARIFFS**High voltage time of use demand network with ActewAGL low voltage network****111 HV TOU Demand Network**

Network access per connection point	\$/day	19.287	0.000	0.000	19.287
Maximum demand charge	c/KVA/day	9.779	7.173	0.000	16.952
Capacity charge	c/KVA/day	9.779	7.173	0.000	16.952
Energy at business times	cents/kWh	1.669	1.653	1.337	4.659
Energy at evening times	cents/kWh	1.097	0.591	1.012	2.700
Energy at off-peak times	cents/kWh	0.386	0.017	0.724	1.127

112 HV TOU Demand Network – Customer HV

Network access per connection point	\$/day	19.287	0.000	0.000	19.287
Maximum demand charge	c/KVA/day	8.764	7.173	0.000	15.937
Capacity charge	c/KVA/day	8.764	7.173	0.000	15.937
Energy at business times	cents/kWh	1.669	1.653	1.337	4.659
Energy at evening times	cents/kWh	1.097	0.591	1.012	2.700
Energy at off-peak times	cents/kWh	0.386	0.017	0.724	1.127

High voltage time of use demand network without ActewAGL low voltage network**121 HV TOU Demand Network – Customer LV**

Network access per connection point	\$/day	19.287	0.000	0.000	19.287
Maximum demand charge	c/KVA/day	9.779	7.173	0.000	16.952
Capacity charge	c/KVA/day	9.779	7.173	0.000	16.952
Energy at business times	cents/kWh	1.267	1.649	1.337	4.253
Energy at evening times	cents/kWh	0.746	0.588	1.012	2.346
Energy at off-peak times	cents/kWh	0.245	0.015	0.724	0.984

122 HV TOU Demand Network – Customer HV and LV

Network access per connection point	\$/day	19.287	0.000	0.000	19.287
Maximum demand charge	c/KVA/day	12.400	3.537	0.000	15.937
Capacity charge	c/KVA/day	12.400	3.537	0.000	15.937
Energy at business times	cents/kWh	2.098	0.818	1.337	4.253
Energy at evening times	cents/kWh	1.043	0.290	1.012	2.345

Energy at off-peak times	cents/kWh	0.254	0.006	0.724	0.984
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2.4 Changes in network charges in 2016/17

Network charges will increase by 1.5 per cent, on average, in 2016/17 in line with CPI, as per the Undertaking.

Table 2-5 shows the AER approved network charges for 2015/16 and 2016/17, excluding GST. The non-capital meter charges have been included. The table also shows the amount of the average change in prices and the average percentage change in prices.

High voltage charges do not include metering capital charges as ActewAGL Distribution does not provide metering services to these customers.

Table 2-5 Change in network use of system charges 2015/16 to 2016/17 (including non-capital metering)

Description	Unit	Network & metering charges 2015/16	Network & metering charges 2016/17	Average Change c/kWh	Average Change %
RESIDENTIAL TARIFFS					
10 Residential Basic Network					
Network access charge	cents/day	36.50	37.05	0.13	1.5%
Energy consumption	cents/kWh	6.80	6.90		
15 Residential TOU Network					
Network access charge	cents/day	36.50	37.05	0.14	1.5%
Energy at max times	cents/kWh	11.76	11.94		
Energy at mid times	cents/kWh	5.68	5.77		
Energy at economy times	cents/kWh	2.57	2.61		
20 Residential 5000 Network					
Network access charge	cents/day	57.70	58.57	0.13	1.5%
Energy for the first 60 kWh per day	cents/kWh	5.28	5.36		
Energy above 60 kWh per day	cents/kWh	6.80	6.90		
30 Residential with Heat Pump Network					
Network access charge	cents/day	100.40	101.92	0.10	1.5%
Energy for the first 165 kWh per day	cents/kWh	3.83	3.89		
Energy above 165 kWh per day	cents/kWh	6.80	6.90		
60 Off-Peak (1) Night Network					
Energy consumption	cents/kWh	1.85	1.88	0.03	1.5%
70 Off-Peak (3) Day & Night Network					
Energy consumption	cents/kWh	2.73	2.77	0.04	1.5%
Renewable Energy Generation					
Gross metered energy	cents/kWh	0.00	0.00	0.00	
COMMERCIAL LOW VOLTAGE TARIFFS					
40 General Network					
Network access charge	cents/day	66.00	67.00	0.18	1.5%
Energy for the first 330 kWh per day	cents/kWh	10.57	10.73		
Energy above 330 kWh per day	cents/kWh	13.75	13.96		
135 Small Unmetered Loads Network					
Network access charge	cents/day	37.70	38.27	0.17	1.5%

Energy consumption	cents/kWh	11.041	11.207		
80 Streetlighting Network				0.11	1.5%
Network access charge	cents/day	60.00	60.91		
Energy consumption	cents/kWh	7.21	7.32		
90 General TOU Network				0.15	1.5%
Network access charge	cents/day	66.00	67.00		
Energy at business times	cents/kWh	16.71	16.96		
Energy at evening times	cents/kWh	8.20	8.32		
Energy at off-peak times	cents/kWh	3.39	3.44		
Low voltage time of use demand network					
101 LV TOU kVA Demand Network				0.12	1.5%
Network access per connection point	cents/day	157.12	159.49		
Maximum demand charge	c/KVA/day	41.70	42.33		
Energy at business times	cents/kWh	6.43	6.53		
Energy at evening times	cents/kWh	3.43	3.48		
Energy at off-peak times	cents/kWh	1.54	1.56		
103 LV TOU Capacity Network				0.10	1.5%
Network access per connection point	cents/day	157.12	159.49		
Maximum demand charge	c/KVA/day	19.50	19.79		
Capacity charge	c/KVA/day	19.50	19.79		
Energy at business times	cents/kWh	6.43	6.53		
Energy at evening times	cents/kWh	3.43	3.48		
Energy at off-peak times	cents/kWh	1.54	1.56		
HIGH VOLTAGE TARIFFS					
High voltage time of use demand network with ActewAGL LV network					
111 HV TOU Demand Network				0.08	1.5%
Network access per connection point	\$/day	19.00	19.29		
Maximum demand charge	c/KVA/day	16.70	16.95		
Capacity charge	c/KVA/day	16.70	16.95		
Energy at business times	cents/kWh	4.59	4.66		
Energy at evening times	cents/kWh	2.66	2.70		
Energy at off-peak times	cents/kWh	1.11	1.13		
112 HV TOU Demand Network – Customer HV					
Network access per connection point	\$/day	19.00	19.82		
Maximum demand charge	c/KVA/day	15.70	15.94		
Capacity charge	c/KVA/day	15.70	15.94		
Energy at business times	cents/kWh	4.59	4.66		
Energy at evening times	cents/kWh	2.66	2.70		
Energy at off-peak times	cents/kWh	1.11	1.13		
High voltage time of use demand network without ActewAGL LV network					
121 HV TOU Demand Network – Customer LV				0.08	1.5%
Network access per connection point	\$/day	19.00	19.82		
Maximum demand charge	c/KVA/day	16.70	16.95		
Capacity charge	c/KVA/day	16.70	16.95		
Energy at business times	cents/kWh	4.19	4.25		
Energy at evening times	cents/kWh	2.31	2.35		
Energy at off-peak times	cents/kWh	0.97	0.98		
122 HV TOU Demand Network – Customer HV and LV				0.09	1.6%
Network access per connection point	\$/day	19.00	19.82		
Maximum demand charge	c/KVA/day	15.70	15.94		
Capacity charge	c/KVA/day	15.70	15.94		
Energy at business times	cents/kWh	4.19	4.25		
Energy at evening times	cents/kWh	2.31	2.34		

3. Charges for metering and ancillary services

3.1 Ancillary services

Table 3-1 shows ActewAGL Distribution's approved charges for ancillary services and its connection service charges for 2016/17, excluding and including GST. The 2016/17 charges have been set as per the Undertaking, and consistent with the AER's Final Decision.

Table 3-1 Charges for ancillary and connection services 2016/17

Code	Service	Unit	Prices excl GST 2016/17	Prices incl GST 2016/17
Premise Re-energisation – Existing Network Connection				
501	Re-energise premise – Business Hours	per visit	67.82	74.60
502	Re-energise premise – After Hours	per visit	85.97	94.57
Premise De-energisation – Existing Network Connection				
503	De-energise premise – Business Hours	per visit	67.82	74.60
505	De-energise premise for debt non-payment	per test	135.65	149.22
Meter Reconfiguration				
507	Install single phase, single element manually read interval meter	per meter	511.93	563.12
508	Install subsequent single phase, single element meter - same location & visit	per meter	323.64	356.00
509	Install single phase, two element meter	per meter	622.57	684.83
511	Install subsequent single phase, two element meter - same location & visit	per meter	434.28	477.71
512	Install three phase meter	per meter	749.64	824.60
513	Install subsequent three phase meter - same location & visit	per meter	561.34	617.47
Meter Investigations				
504	Meter Test (Whole Current) – Business Hours	per test	271.30	298.43
510	Meter Test (CT/VT) – Business Hours	per test	314.20	345.62
Special Meter Services				
506	Special Meter Read	per read	31.37	34.51
Temporary Network Connections (excludes meter cost)				
520	Temporary Builders Supply – Overhead (Business Hours) (excludes meter cost)	per installation	609.61	670.57
522	Temporary Builders Supply – Underground (Business Hours) (excludes meter costs)	per installation	1,330.82	1,463.90

New Network Connections (excludes meter installation)				
523	New Underground Service Connection – Greenfield	per installation	-	-
526	New Overhead Service Connection – Brownfield (Business Hours)	per installation	800.66	880.73
527	New Underground Service Connection – Brownfield from Front	per installation	1,330.82	1,463.90
528	New Underground Service Connection – Brownfield from Rear	per installation	1,330.82	1,463.90
Network Connection Alterations and Additions (excludes meter installation)				
541	Overhead Service Relocation – Single Visit (Business Hours)	per installation	764.19	840.61
542	Overhead Service Relocation – Two Visits (Business Hours)	per installation	1,528.37	1,681.21
543	Overhead Service Upgrade – Service Cable Replacement Not Required	per installation	764.19	840.61
544	Overhead Service Upgrade – Service Cable Replacement Required	per installation	800.66	880.73
545	Underground Service Upgrade – Service Cable Replacement Not Required	per installation	1,294.36	1,423.80
546	Underground Service Upgrade – Service Cable Replacement Required	per installation	1,330.82	1,463.90
547	Underground Service Relocation – Single Visit (Business Hours)	per installation	1,330.82	1,463.90
548	Install surface mounted point of entry (POE) box	per installation	615.47	677.02
Temporary De-energisation				
560	Temporary de-energisation – LV (Business Hours)	per occurrence	406.95	447.65
561	Temporary de-energisation – HV (Business Hours)	per occurrence	406.95	447.65
Supply Abolishment / Removal				
562	Supply Abolishment / Removal – Overhead (Business Hours)	per site visit	573.15	630.47
563	Supply Abolishment / Removal - Underground (Business Hours)	per site visit	1,035.49	1,139.04
Miscellaneous Customer Initiated Services				
564	Install & Remove Tiger Tails – Per Installation (Business Hours)	per installation	1,345.92	1,480.51
565	Install & Remove Tiger Tails - Per Span (Business Hours)	per installation	677.55	745.31
566	Install & Remove Warning Flags – Per Installation (Business Hours)	per installation	1,146.28	1,260.91
567	Install & Remove Warning Flags - Per Span (Business Hours)	per installation	580.75	638.83
Embedded Generation - Operational & Maintenance Fees				
568	Small Embedded Generation OPEX Fees - Connection Assets	per annum	2%	2%
569	Small Embedded Generation OPEX Fees - Shared Network Asset	per annum	2%	2%
Connection Enquiry Processing - PV Installations				
570	PV Connection Enquiry – LV Class 1 (<= 10kW Single Phase / 30kW Three Phase)	per installation	-	-
571	PV Connection Enquiry – LV Class 2 to 5 (> 30kW <= 1500kW Three Phase)	per installation	557.20	612.92
572	PV Connection Enquiry – HV	per installation	1,114.41	1,225.85
573	Provision of information for Network technical study for large scale installations	per installation	11,144.12	12,258.53

Network Design & Investigation / Analysis Services - PV Installations

(See:

<http://www.actewagl.com.au/~media/ActewAGL/ActewAGL-Files/About-us/Publications/Guidelines-for-LV-embedded-generator-connections.ashx>)

574	Design & Investigation - LV Connection Class 1 PV (<= 10kW Single Phase / 30kW Three Phase)	per installation		
575	Design & Investigation - LV Connection Class 2 PV (> 30kW and <= 60kW Three Phase)	per installation	3,714.71	4,086.18
576	Design & Investigation - LV Connection Class 3 PV (> 60 kW and <= 120kW Three Phase)	per installation	5,572.05	6,129.26
577	Design & Investigation - LV Connection Class 4 PV (> 120 kW and <= 200kW Three Phase)	per installation	7,429.41	8,172.35
578	Design & Investigation - LV Connection Class 5 PV (> 200kW and <= 1500kW Three Phase) – ActewAGL Network Study	per installation	11,144.12	12,258.53
579	Design & Investigation - HV Connection Class 5 PV (> 200kW and <= 1500kW Three Phase) – Customer Network Study	per installation	13,930.14	15,323.15
Residential Estate Subdivision Services				
580	URD Subdivision Electricity Distribution Network Reticulation - Multi-Unit Blocks	per block	0.00	0.00
581	URD Subdivision Electricity Distribution Network Reticulation - Blocks <= 650 m2	per block	1,678.95	1,846.85
582	URD Subdivision Electricity Distribution Network Reticulation - Blocks 650 - 1100m2 with average linear frontage of 22-25 meters	per block	2,199.69	2,419.66
Upstream Augmentation				
585	HV Feeder	per KVA	36.37	40.01
586	Distribution substation	per KVA	21.06	23.17
Rescheduled Site Visits				
590	Rescheduled Site Visit – One Person	per site visit	135.65	149.22
591	Rescheduled Site Visit – Service Team	per site visit	573.15	630.47
Trenching charges				
592	Trenching - first 2 meters	per visit	520.26	572.29
593	Trenching - subsequent meters	per meter	120.99	133.09
Boring charges				
594	Under footpath	per occurrence	943.73	1,038.10
595	Under driveway	per occurrence	1,125.22	1,237.74

3.2 Metering services charges for 2016/17

In 2015/16, there were a series of changes made to metering service charges. In its Final Decision, the AER approved two types of metering service charges:

- Upfront capital charge (for all new and upgraded meters installed from 1 July 2015); and
- Annual charge comprising two components:
 - o capital —metering asset base recovery; and
 - o non-capital —operating expenditure and tax.

These changes remain in place in 2016/17.

For existing regulated meters installed before 30 June 2015, ActewAGL Distribution has paid upfront for the capital costs of these meters which were then added to the asset base and recovered gradually, over the life of the meter, through annual charges. These customers pay the following charges:

- Capital component of regulated annual metering charge
- Non-capital component of the regulated annual metering charge.

For regulated new meter connections installed after 1 July 2015, the capital costs of the meter are paid upfront by the customer. As they have already paid for their capital component upfront, the only costs relating to their regulated metering service left to be recovered through annual charges are the non-capital costs.

To facilitate records of these customers, ActewAGL Distribution has network tariffs that exclude metering capital charges (XMC tariffs). These network tariffs are applied to new connections that have paid for their metering assets. The unmetered loads do not have an XMC tariff because ActewAGL Distribution has not connected meters to these loads. Also, the off-peak network tariffs do not have an equivalent XMC tariff because the metering costs are associated with the customer's substantive tariff, not the supplementary off-peak tariff. Furthermore, there are no high voltage XMC network tariffs, because high voltage network tariffs exclude metering charges as ActewAGL Distribution has not provided manually read meters to these customers since they have been required to use remotely read (types 1- 4) meters. The XMC tariffs ensure that ActewAGL Distribution and Retailers are able to clearly identify, through the network billing system, which customers have paid for their meters and are therefore not liable for the metering capital charge.

3.3 Metering non-capital charges for 2016/17

Table 3-2 presents the metering non-capital charges for 2016/17. The annual metering non-capital charges apply to both existing and new metering customers. The non-capital metering charges in 2016/17 are 1.5 per cent higher than in 2015/16, as per the Enforceable Undertaking.

Table 3-2 Metering non-capital charges, 2016/17

Code	Description	Unit	Excl. GST 2016/17	Incl. GST 2016/17
MP1	Quarterly basic metering rate Accumulation and time-of-use meters read quarterly	cents per day per NMI*	3.766	4.143
MP2	Monthly basic metering rate Accumulation and time-of-use meters read monthly	cents per day per NMI	6.588	7.247
MP3	Time-of-use metering rate Time-of-use meters read monthly	cents per day per NMI	6.588	7.247
MP4	Monthly manually-read interval metering rate Interval meters recording at either 15- or 30-minute intervals, read manually and processed monthly	cents per day per NMI	53.190	58.509
MP6	Quarterly manually-read interval metering rate Interval meters recording at either 15- or 30-minute intervals, read manually and processed quarterly	cents per day per NMI	15.176	16.694

*National Meter Identifier.

3.4 Metering capital charges for 2016/17

The metering capital charges for 2016/17 are shown below in Table 3-3 and were added to the network charges in Table 2-4. These amounts are derived by applying CPI of 1.5 per cent to the 2015/16 metering capital charges, as per the Undertaking.

Table 3-3 Metering capital charges, 2016/17

Code	Description	Unit	Excluding GST 2016/17	Including GST 2016/17
MP7	Quarterly basic metering capital rate Accumulation and time-of-use meters read quarterly	cents per day per NMI*	7.644	8.408
MP8	Monthly basic metering capital rate Accumulation and time-of-use meters read monthly	cents per day per NMI	13.369	14.706
MP9	Time-of-use metering capital rate Time-of-use meters read monthly	cents per day per NMI	13.369	14.706
MP10	Monthly manually-read interval metering capital rate Interval meters recording at either 15- or 30-minute intervals, read manually and processed monthly	cents per day per NMI	107.903	118.693

*National Meter Identifier

4. Expected price trends

4.1 Expected distribution price trends

Table 4-1 presents ActewAGL Distribution's indicative distribution charges for 2017/18 and 2018/19.

As per the Undertaking, distribution charges for 2016/17 have been set based on the 2015/16 distribution charges, adjusted for changes in the CPI. The CPI used to calculate the 2016/17 charges is calculated in a manner that is consistent with the formulae for distribution standard control services set out in the Final Determination (Attachment 14 – Figure 14.1).

To be consistent, indicative distribution prices in 2017/18 and 2018/19 have also been calculated by adjusting the charges from the previous year by CPI. The CPI used to calculate the 2017/18 and 2018/19 prices is 2.38 per cent, as approved by the AER in its Final Determination. While the Final Determination has been set aside by the Tribunal and the AER has applied to the Federal Court for a judicial review of the Tribunal's decision, the effect of this delay has created uncertainty for users about applicable prices. The Undertaking, proffered by AAD to ameliorate this uncertainty, states that the CPI formulae in the Final Decision is to be used as the basis for the 2016/17 prices, and is therefore also relevant when estimating 2017/18 and 2018/19 indicative prices.

Table 4-1 Indicative distribution use of system charges 2017/18 to 2018/19 (excluding GST)

Code	Description	Unit	2017/18	2018/19
10	Residential Basic Network			
	Network access charge	cents/day	26.25	26.88
	Energy consumption	cents/kWh	3.57	3.65
15	Residential TOU Network			
	Network access charge	cents/day	26.25	26.88
	Energy at max times	cents/kWh	9.38	9.60
	Energy at mid times	cents/kWh	3.80	3.89
	Energy at economy times	cents/kWh	1.15	1.18
20	Residential 5000 Network			
	Network access charge	cents/day	48.28	49.43
	Energy for the first 60 kWh per day	cents/kWh	2.23	2.29
	Energy above 60 kWh per day	cents/kWh	3.57	3.65
30	Residential with Heat Pump Network			
	Network access charge	cents/day	92.66	94.86
	Energy for the first 165 kWh per day	cents/kWh	0.74	0.76
	Energy above 165 kWh per day	cents/kWh	3.57	3.65

40	General Network			
	Network access charge	cents/day	48.16	49.30
	Energy for the first 330 kWh per day	cents/kWh	6.50	6.66
	Energy above 330 kWh per day	cents/kWh	9.77	10.01
60	Off-Peak (1) Night Network			
	Energy consumption	cents/kWh	0.08	0.08
70	Off-Peak (3) Day & Night Network			
	Energy consumption	cents/kWh	0.36	0.37
80	Streetlighting Network			
	Network access charge	cents/day	48.67	49.83
	Energy consumption	cents/kWh	5.43	5.56
90	General TOU Network			
	Network access charge	cents/day	48.16	49.30
	Energy at business times	cents/kWh	13.72	14.04
	Energy at evening times	cents/kWh	6.30	6.45
	Energy at off-peak times	cents/kWh	2.54	2.60
	Low voltage time of use demand network			
101	LV TOU kVA Demand Network			
	Network access per connection point	cents/day	52.27	53.52
	Maximum demand charge	c/KVA/day	29.96	30.67
	Energy at business times	cents/kWh	2.64	2.71
	Energy at evening times	cents/kWh	1.83	1.88
	Energy at off-peak times	cents/kWh	0.80	0.82
103	LV TOU Capacity Network			
	Network access per connection point	cents/day	52.27	53.52
	Maximum demand charge	c/KVA/day	17.13	17.53
	Capacity charge	c/KVA/day	17.13	17.53
	Energy at business times	cents/kWh	3.98	4.07
	Energy at evening times	cents/kWh	2.18	2.23
	Energy at off-peak times	cents/kWh	0.83	0.85
111	HV TOU Demand Network			
	Network access per connection point	\$/day	19.75	20.22
	Maximum demand charge	c/KVA/day	10.01	10.25
	Capacity charge	c/KVA/day	10.01	10.25
	Energy at business times	cents/kWh	1.71	1.75
	Energy at evening times	cents/kWh	1.12	1.15
	Energy at off-peak times	cents/kWh	0.40	0.40
112	HV TOU Demand Network – Customer HV			
	Network access per connection point	\$/day	19.75	20.22
	Maximum demand charge	c/KVA/day	8.97	9.19
	Capacity charge	c/KVA/day	8.97	9.19
	Energy at business times	cents/kWh	1.71	1.75
	Energy at evening times	cents/kWh	1.12	1.15
	Energy at off-peak times	cents/kWh	0.40	0.40

121	HV TOU Demand Network – Customer LV			
	Network access per connection point	\$/day	19.75	20.22
	Maximum demand charge	c/KVA/day	10.01	10.25
	Capacity charge	c/KVA/day	10.01	10.25
	Energy at business times	cents/kWh	1.30	1.33
	Energy at evening times	cents/kWh	0.76	0.78
	Energy at off-peak times	cents/kWh	0.25	0.26
122	HV TOU Demand Network – Customer HV and LV			
	Network access per connection point	\$/day	19.75	20.22
	Maximum demand charge	c/KVA/day	12.70	13.00
	Capacity charge	c/KVA/day	12.70	13.00
	Energy at business times	cents/kWh	2.15	2.20
	Energy at evening times	cents/kWh	1.07	1.09
	Energy at off-peak times	cents/kWh	0.26	0.27
135	Small Unmetered Loads Network			
	Network access charge	cents/day	39.18	40.11
	Energy consumption	cents/kWh	9.50	9.72

4.2 Expected metering price trends

Indicative metering capital and non-capital charges for 2017/18 and 2018/19 are shown in Table 4-2.¹⁴

Table 4-2 Indicative metering charges 2017/18 and 2018/19 (excluding GST)

Code	Description	Unit	2017/18	2018/19
MP1	Quarterly basic metering rate			
	Accumulation and time-of-use meters read quarterly	cents per day per NMI*	3.9	3.9
MP2	Monthly basic metering rate			
	Accumulation and time-of-use meters read monthly	cents per day per NMI	6.7	6.9
MP3	Time-of-use metering rate			
	Time-of-use meters read monthly	cents per day per NMI	6.7	6.9
MP4	Monthly manually-read interval metering rate			
	Interval meters recording at either 15- or 30-minute intervals, read manually and processed monthly	cents per day per NMI	54.5	55.8
MP6	Quarterly manually-read interval metering rate			
	Interval meters recording at either 15- or 30-minute intervals, read manually and processed quarterly	cents per day per NMI	15.5	15.9
MP7	Quarterly basic metering capital rate			
	Accumulation and time-of-use meters read quarterly	cents per day per NMI*	7.8	8.0
MP8	Monthly basic metering capital rate			
	Accumulation and time-of-use meters read monthly	cents per day per NMI	13.7	14.0
MP9	Time-of-use metering capital rate			
	Time-of-use meters read monthly	cents per day per NMI	13.7	14.0
MP10	Monthly manually-read interval metering capital rate			
	Interval meters recording at either 15- or 30-minute intervals, read manually and processed monthly	cents per day per NMI	110.5	113.1

¹⁴ The indicative prices in Table 4-2 do not include an X factor. They are calculated by escalating the charges from the previous year by a CPI of 2.38%.

4.3 Expected ancillary and connection service price trends

Indicative charges for ancillary services for 2017/18 and 2018/19 are shown in Table 4-3. These prices are based on the AER's CPI assumption of 2.38 per cent per annum¹⁵.

Table 4-3 Indicative ancillary & connection service charges (excluding GST)

Code	Service	Unit	2017/18	2018/19
Premise Re-energisation – Existing Network Connection				
501	Re-energise premise – Business Hours	per visit	69.43	71.09
502	Re-energise premise – After Hours	per visit	88.02	90.11
Premise De-energisation – Existing Network Connection				
503	De-energise premise – Business Hours	per visit	69.43	71.09
505	De-energise premise for debt non-payment	per test	138.88	142.18
Meter Reconfiguration				
507	Install single phase, single element manually read interval meter	per meter	524.11	536.59
508	Install subsequent single phase, single element meter - same location & visit	per meter	331.34	339.23
509	Install single phase, two element meter	per meter	637.39	652.56
511	Install subsequent single phase, two element meter - same location & visit	per meter	444.62	455.20
512	Install three phase meter	per meter	767.48	785.75
513	Install subsequent three phase meter - same location & visit	per meter	574.70	588.38
Meter Investigations				
504	Meter Test (Whole Current) – Business Hours	per test	277.76	284.37
510	Meter Test (CT/VT) – Business Hours	per test	321.68	329.33
Special Meter Services				
506	Special Meter Read	per read	32.12	32.88
Temporary Network Connections (excludes meter cost)				
520	Temporary Builders Supply – Overhead (Business Hours) (excludes meter cost)	per installation	624.12	638.97
522	Temporary Builders Supply – Underground (Business Hours) (excludes meter costs)	per installation	1362.49	1394.92
New Network Connections (excludes meter installation)				
523	New Underground Service Connection – Greenfield	per installation	-	-
526	New Overhead Service Connection – Brownfield (Business Hours)	per installation	819.72	839.22
527	New Underground Service Connection – Brownfield from Front	per installation	1362.49	1394.92
528	New Underground Service Connection – Brownfield from Rear	per installation	1362.49	1394.92
Network Connection Alterations and Additions (excludes meter installation)				
541	Overhead Service Relocation – Single Visit (Business Hours)	per installation	782.38	801.00
542	Overhead Service Relocation – Two Visits (Business Hours)	per installation	1564.75	1601.99
543	Overhead Service Upgrade – Service Cable Replacement Not Required	per installation	782.38	801.00
544	Overhead Service Upgrade – Service Cable Replacement Required	per installation	819.72	839.22
545	Underground Service Upgrade – Service Cable Replacement Not Required	per installation	1325.17	1356.70
546	Underground Service Upgrade – Service Cable Replacement	per	1362.49	1394.92

¹⁵ The indicative prices in Table 4-3 do not include an X factor. They are calculated by escalating the charges from the previous year by a CPI of 2.38%.

	Required	installation		
547	Underground Service Relocation – Single Visit (Business Hours)	per installation	1362.49	1394.92
548	Install surface mounted point of entry (POE) box	per installation	630.12	645.11
Temporary De-energisation				
560	Temporary de-energisation – LV (Business Hours)	per occurrence	416.64	426.55
561	Temporary de-energisation – HV (Business Hours)	per occurrence	416.64	426.55
Supply Abolishment / Removal				
562	Supply Abolishment / Removal – Overhead (Business Hours)	per site visit	586.79	600.76
563	Supply Abolishment / Removal - Underground (Business Hours)	per site visit	1060.13	1085.37
Miscellaneous Customer Initiated Services				
564	Install & Remove Tiger Tails – Per Installation (Business Hours)	per installation	1377.95	1410.75
565	Install & Remove Tiger Tails - Per Span (Business Hours)	per installation	693.68	710.19
566	Install & Remove Warning Flags – Per Installation (Business Hours)	per installation	1173.56	1201.49
567	Install & Remove Warning Flags - Per Span (Business Hours)	per installation	594.57	608.72
Embedded Generation - Operational & Maintenance Fees				
568	Small Embedded Generation OPEX Fees - Connection Assets	per annum	2%	2%
569	Small Embedded Generation OPEX Fees - Shared Network Asset	per annum	2%	2%
Connection Enquiry Processing - PV Installations				
570	PV Connection Enquiry – LV Class 1 (<= 10kW Single Phase / 30kW Three Phase)	per installation	-	-
571	PV Connection Enquiry – LV Class 2 to 5 (> 30kW <= 1500kW Three Phase)	per installation	570.46	584.04
572	PV Connection Enquiry – HV	per installation	1140.93	1168.09
573	Provision of information for Network technical study for large scale installations	per installation	11409.35	11680.89
Network Design & Investigation / Analysis Services - PV Installations				
(See: http://www.actewagl.com.au/~media/ActewAGL/ActewAGL-Files/About-us/Publications/Guidelines-for-LV-embedded-generator-connections.ashx)				
574	Design & Investigation - LV Connection Class 1 PV (<= 10kW Single Phase / 30kW Three Phase)	per installation	-	-
575	Design & Investigation - LV Connection Class 2 PV (> 30kW and <= 60kW Three Phase)	per installation	3803.12	3893.63
576	Design & Investigation - LV Connection Class 3 PV (> 60 kW and <= 120kW Three Phase)	per installation	5704.66	5840.44
577	Design & Investigation - LV Connection Class 4 PV (> 120 kW and <= 200kW Three Phase)	per installation	7606.23	7787.26
578	Design & Investigation - LV Connection Class 5 PV (> 200kW and <= 1500kW Three Phase) – ActewAGL Network Study	per installation	11409.35	11680.89
579	Design & Investigation - HV Connection Class 5 PV (> 200kW and <= 1500kW Three Phase) – Customer Network Study	per installation	14261.68	14601.11
Residential Estate Subdivision Services				
580	URD Subdivision Electricity Distribution Network Reticulation - Multi-Unit Blocks	per block	-	-
581	URD Subdivision Electricity Distribution Network Reticulation - Blocks <= 650 m2	per block	1718.91	1759.82
582	URD Subdivision Electricity Distribution Network Reticulation - Blocks 650 - 1100m2 with average linear frontage of 22-25 meters	per block	2252.04	2305.64
Upstream Augmentation				
585	HV Feeder	per KVA	37.24	38.12
586	Distribution substation	per KVA	21.56	22.07

Rescheduled Site Visits				
590	Rescheduled Site Visit – One Person	per site visit	138.88	142.18
591	Rescheduled Site Visit – Service Team	per site visit	586.79	600.76
Trenching charges				
592	Trenching - first 2 meters	per visit	532.64	545.32
593	Trenching - subsequent meters	per meter	123.87	126.82
Boring charges				
594	Under footpath	per occurrence	966.19	989.19
595	Under driveway	per occurrence	1152.00	1179.42